IronHack - AI Consulting & Integration

LAB M1.02 | Customer Churn Prediction

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1. Summary of approach

I built a K-Nearest Neighbors (KNN) model to predict which customers of a telecom company are likely to leave (churn). I used the Telco Customer Churn dataset: 7,043 customers and 21 columns.

1. **Load and explore:** checked the shape, data types, missing values and the target distribution. 26.5% of customers churned.
2. **Preprocess:**
    - `TotalCharges` was stored as text and hid 11 blank values. I converted it to numbers and dropped those 11 rows (7,032 left).
    - Dropped `customerID`, which identifies customers but has no predictive value.
    - Mapped the target `Churn` to 1 (left) and 0 (stayed).
    - Converted categorical columns to numbers with one-hot encoding (`pd.get_dummies`), giving 30 features.
3. **Split:** 80% training (5,625 customers) and 20% test (1,407), with `stratify` to keep the churn rate equal in both sets and `random_state=42` for reproducibility.
4. **Train and evaluate:** trained KNN with K=5, then measured accuracy, precision, recall and the confusion matrix on the test set.
5. **Experiment:** compared several K values and chose the best one.

2. Model performance

The best model (K=7) reached 76.6% accuracy but caught only 45% of the customers who churned. Because only 26.5% of customers churn, a "model" that always predicts *no churn* is already 73.4% accurate on the test set. That is the **baseline** any real model must beat, so accuracy alone is misleading here.

Model	Accuracy	Precision (churn)	Recall (churn)
Baseline (always "no churn")	73.4%	-	0%
KNN, K=3	73.8%	50.9%	44.1%
KNN, K=5	74.6%	52.8%	42.8%
KNN, K=7 (chosen)	**76.6%**	**57.7%**	**45.2%**
KNN, K=9	76.2%	57.4%	40.4%

K=7 was the best value on all three metrics, so no trade-off was needed. A larger K (9) lowered recall: the vote leans toward the majority class, so fewer churners are caught.

Confusion matrix (K=5, test set): of 374 customers who churned, the model found 160 and missed 214. Of 1,033 who stayed, it flagged 143 by mistake (false alarms).

- **Precision** (~58% at K=7): when the model flags a customer as at risk, it is right a little more than half the time.
- **Recall** (~45%): the model misses more than half of the real churners, which is the most important weakness for the business.

3. Key findings about customer churn

Customers on flexible contracts, new customers and fiber optic users churn the most. We measured each feature's correlation with churn (from -1 to +1; further from 0 means a stronger link).

Feature	Correlation with churn	Meaning
Tenure (months as a customer)	-0.35	Longer customers stay; churners average 18 months vs 38 for those who stay
Fiber optic internet	+0.31	Fiber users churn much more than DSL users
Two-year contract	-0.30	Long contracts strongly protect against churn; month-to-month customers churn the most
Electronic check payment	+0.30	Customers paying by electronic check churn more than those on automatic payment
Monthly charges	+0.19	Churners pay more per month (about 74 vs 61 on average)
Paperless billing	+0.19	Linked to higher churn
Senior citizen	+0.15	Older customers churn somewhat more

Month-to-month contract does not appear in the correlation list because one-hot encoding (`drop_first=True`) used it as the reference category. Its effect shows through the two-year contract result.

4. Business recommendations

The company should focus retention on new, month-to-month and fiber optic customers, and push them toward longer contracts.

1. **Target the high-risk groups:** month-to-month customers, customers in their first months, and fiber optic users.
2. **Reward longer contracts:** offer discounts or benefits for moving to a one- or two-year contract, the strongest protection against churn.
3. **Review the fiber optic offer:** check whether price or service quality drives these customers away.
4. **Encourage automatic payments:** electronic-check customers churn more; an incentive to switch to bank transfer or card could help.
5. **Use the model as an early-warning list:** contact flagged customers with a retention offer. Some false alarms are acceptable, because a retention offer usually costs less than losing a customer.

5. Limitations and future improvements

The model is only slightly better than the baseline and misses more than half of the churners, so it needs improvement before real use.

Limitations

- **Low recall:** about 55% of customers who leave are not detected.
- **Small gain over the baseline:** 76.6% accuracy vs 73.4% for always predicting "no churn".
- **No explanation:** KNN predicts *who* may churn but not *why*, which limits its use for business decisions.
- **Unscaled features:** KNN uses distances, so large-valued columns (e.g. `TotalCharges`, up to ~8,700) dominate the 0/1 columns.
- **Single split:** results come from one train/test split and could change with a different one.

Future improvements

- Scale the features with `StandardScaler` (fitted on the training data only), which is known to improve KNN.

- Try models that handle many features and explain their decisions, such as Logistic Regression or Random Forest.
- Use cross-validation for more reliable results.
- Handle the class imbalance (e.g. class weights) and adjust the decision threshold to favour recall.